



REVIEW

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A systematic review on GIS-based road traffic accidents analysis and road safety audit

Destaw Kifle Endashaw^{1,2*} , Kidanemariam Alula Habtegiorgis¹ , Baqer Muhammad Al-Ramadan^{3,4} , Hassan Musaed Al-Ahmadi^{1,4} and Basha Fayissa Deressa¹

Abstract

Road traffic accidents (RTAs) cause approximately 1.35 million deaths and 50 million injuries annually, disproportionately affecting people aged 5–29 years. The objective of this review was to synthesize how Geographic Information Systems (GIS) support RTA analysis and road safety audits. Relevant articles were searched in different electronic databases such as Scopus, Web of Science, PubMed, and Google Scholar using predefined terms; after screening and eligibility checks, 75 peer-reviewed studies were included. Dominant techniques included Kernel Density Estimation (KDE), Getis–Ord Gi* clustering, crash rate analysis, and Empirical Bayes (EB) analyses, as well as machine-learning clustering. Across contexts, GIS consistently identified spatial blackspots, supported spatiotemporal trend analysis, and informed targeted countermeasures; key limitations were heterogeneous data quality, inconsistent methodological choices, and the integration of real-time and behavioral data. GIS is effective for blackspot detection and decision support in road safety. Future work should prioritize standardizing methods, incorporating real-time IoT streams and deep learning, and integrating behavioral and exposure data to improve prediction and intervention design.

Keywords Cluster analyses, GIS, Road traffic, Accidents, Spatial analysis, Safety

1 Introduction

Road traffic accidents (RTAs) are a major global public health problem, accounting for 1.35 million deaths and 50 million injuries annually (Berhane G/Yesus, 2015; Satria, 2020). These accidents have devastating physical, mental, and economic impacts on individuals and communities, especially among people aged 5 to 29 years. Increased urbanization, population density, and vehicle usage have increased the demand to address the issues of traffic

safety (Novikov et al., 2021). With the aid of GIS, some of the key advances in spatial analysis have been possible, which have helped transportation authorities to identify accident prone areas, generally known as "blackspots" and recommend site specific safety measures (Aati et al., 2024; Ashraf & Idrisi, 2024; Zakeer et al., 2025). By managing geographic data in traffic crashes, GIS forms a basis for understanding the causes that underpin accidents, such as defects in road design, driver behavior, congestion patterns and environmental factors. Mapping and visualization of data enable the development of evidence based strategies to reduce the risks and improve safety on roads (Berhane G/Yesus, 2015; Fayaz et al., 2022). Therefore, it has emerged that GIS is a very important tool for the analysis in RTAs and improvements in transportation safety (Ashraf et al., 2024, 2025; Bhele et al., 2024; Munasinghe, 2023).

Over the past decade, there has been an upward surge in applying GIS within traffic safety management. Its ease and flexibility in dealing with big datasets, and

*Correspondence:

Destaw Kifle Endashaw
g202416220@kfupm.edu.sa

¹ Department of Civil and Environmental Engineering, College of Design and Built Environment, King Fahd University of Petroleum and Minerals (KFUPM), Dhahran 31261, Saudi Arabia

² Civil Engineering Department, Gafat Institute of Technology, Debre Tabor University, Debre Tabor, Ethiopia

³ Architecture and City Design Department, King Fahd University of Petroleum and Mineral, Dhahran 31261, Saudi Arabia

⁴ Interdisciplinary Research Center for Smart Mobility and Logistics, King Fahd University of Petroleum and Minerals, Dhahran 31261, Saudi Arabia

conducting complex analyses have made it an indispensable tool to identify high-risk locations, assess crash severity and model the frequency of crashes (Erdogan et al., 2008). Additionally, GIS enables transportation engineers (Fayaz et al., 2022; Sayed & Mhaske, 2013), law enforcement agencies (Berhane G/Yesus, 2015), urban planners (Ordjević, 2024), and public health specialists (Alsahfi, 2024) to integrate and visualize all data for joint analysis on one single platform.

Key applications of GIS include spatial analysis, predictive modeling, and real time monitoring to proactively address traffic safety concerns (Ashraf et al., 2025; Nayak & Goyal, 2024; Science et al., 2022). The application of various GIS tools such as density, proximity, cluster, and pattern analyses play a crucial role in the discovery of spatial relationships and trends that inform decisions related to infrastructure improvements, enforcement of traffic laws, and public safety campaigns (Mansuri et al., 2015). By merging crash data with geographic information, GIS thus places comprehensive and effective intervention designs in the hands of stakeholders to save lives, and prevent injuries (Satria, 2020).

In other words, GIS provides a strong platform for better understanding, analyzing, and solving complexities related to traffic accidents (Ashraf et al., 2025; Djalante, 2020). If applied to planning for traffic safety, it helps identify high-risk locations, motivates coordination of efforts in policies related to safety in traffic, and applies targeted solutions for improved traffic safety worldwide (Ashraf et al., 2024; Fayaz et al., 2022; Nayak & Goyal, 2024). This review paper examines the application of GIS techniques in road safety audits and traffic accident analysis. The paper reviews the literature on traffic accident evaluation using GIS by collecting and reviewing previous studies employing GIS techniques. The study compares different GIS-based techniques of crash analysis to observe how they can be practically applied. In addition, it highlights trends, gaps, and challenges in the literature, including dominant methodologies, sources of data, and key findings, as well as addressing the limitations of research. Lastly, the paper provides a preview of future research directions, outlining avenues for advancing GIS-based accident analysis through technology development, integration with AI, and policymaking.

This review is organized as follows: the next section outlines the search strategy and eligibility criteria, followed by a mapping of the included studies by context, data and technique. The subsequent section compares the core GIS methods, after which the paper discusses implications, limitations and directions for future research. Finally, it concludes by synthesizing key findings.

2 Methodology

This paper represents a critical literature review that examines the application of GIS in road traffic accident analysis and road safety audits. The review of available studies was systematically conducted to identify findings and insights related to road traffic accidents and road safeties. To achieve this, a structured five step research methodology was applied, guiding the selection, analysis and synthesis of relevant literature to strengthen the understanding of how GIS can be applied to traffic safety and audits. This approach aims to update research in this specific area and delivers value to policymakers, researchers, and industry stakeholders.

Initially, the scope of the review was set to include studies published between 2009 and 2025 that addressed road traffic accidents and road safety audits, focusing on outcomes such as blackspot identification, accident severity and intervention strategies. Second, a comprehensive search was conducted across four research databases: Scopus, Web of Science, PubMed, and Google Scholar, using general and combined search terms such as “traffic accidents”, “road safety audit”, “GIS and road safety audits” and “GIS and road traffic accident”. Third, the screening phase consisted of duplicate detection and abstract assessment, resulting in the exclusion of irrelevant studies. A review framework, represented in Fig. 1, illustrates this process through a PRISMA flow diagram.

Fourth, journal and conference articles were further filtered using predefined inclusion and exclusion criteria that aligned with the review objectives Table 1. As (Olayode et al., 2023) note, the application of such criteria ensures that only valuable and relevant data are retained. This step included assessing whether studies met requirements such as peer-reviewed status, publication language, keyword relevance, and publication year. Fifth, the eligible studies were synthesized narratively and quantitatively, mapping their contexts and methods. The overall process was designed to provide a reliable foundation for comparing GIS-based approaches, highlighting trends, gaps, and challenges in their application to traffic safety.

Table 1 outlines the key criteria used to determine whether journal and conference articles were suitable for inclusion in this review, thereby ensuring that only relevant and high-quality studies were retained. Using the defined search terms across the selected databases Scopus, Google Scholar, Web of Science, and PubMed, a total of 184 research articles were initially identified. Table 2 summarizes the search keywords applied, including traffic accidents, road safety audit, GIS, and road traffic accident, and GIS and road safety audit. Following systematic screening, only the most relevant studies were retained. First, non-peer-reviewed articles were excluded, leaving 129 articles.

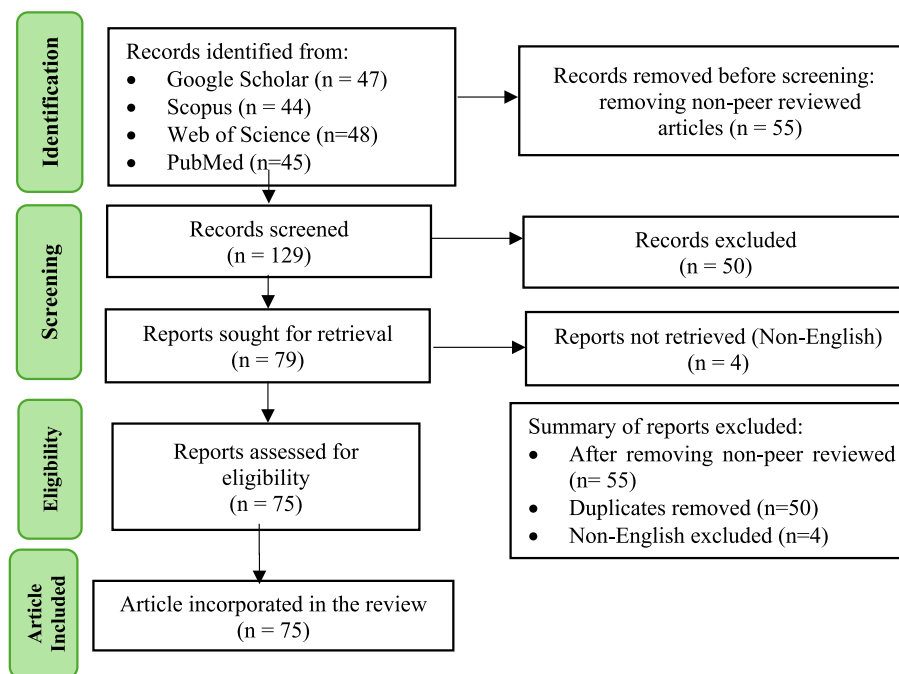


Fig. 1 Flow chart used to identify papers reviewed

Table 1 Inclusion and exclusion criteria for selected articles

Criterion	Include	Exclude	Rationale
Peer-reviewed status	Published in peer-reviewed journals or conference proceedings	Not peer-reviewed (e.g., reports, theses, dissertations)	Ensures scientific reliability and quality control
Language	Written in English	Non-English language	Maintains consistency and accessibility for analysis
Publication year	Published between 2009 and 2025	Published before 2009	Focuses on recent and relevant research developments
Topical relevance	Keywords such as <i>traffic accidents</i> , <i>road safety audit</i> , <i>GIS</i> , and <i>road traffic accident GIS</i> and <i>road safety audit</i> are present in the abstract/keywords	Keywords absent	Guarantees thematic alignment with review objectives
Originality	Unique, innovative articles	Identical/duplicate records	Avoids redundancy and bias in evidence-based
Study type	Empirical, modeling, or review articles using GIS for RTA/RSA	Studies outside the GIS-based RTA or RSA scope	Ensures methodological and topical relevance

Second, duplicate records were removed, resulting in 79 unique studies. Finally, non-English publications were excluded, yielding a total of 75 articles that were included in the final review. Figure 2 demonstrates the analysis of search keywords of how the papers used in the research were reviewed in regard to the four academic databases utilized; namely Scopus, Web of Science, Google Scholar, and PubMed.

2.1 Application of GIS for RTA and RSA

2.1.1 Blackspot detection and spatiotemporal trends

The first step toward improving road safety is the identification of blackspots in a road network or the risk-prone locations, i.e., the blackspot locations (Agoylo, 2024; Berhanu et al., 2023). The theory was first proposed in the early twentieth century that if a person has

Table 2 A summary of selected articles included and excluded article for this review research

Academic Database	Keyword Search	Selected Articles	After Removing Non-peer-reviewed Articles	After Removing Duplicate Articles	Removing Non-English Articles
Google Scholar	Traffic accidents	13	6	5	5
	Road safety audit	8	6	4	4
	GIS and road traffic accident	12	8	4	4
	GIS and road safety audit	14	13	6	6
Scopus	Traffic accidents	8	6	4	4
	Road safety audit	12	9	5	3
	GIS and road traffic accidents	13	10	4	4
	GIS and road safety audit	11	8	5	4
Web of Science	Traffic accidents	12	8	4	4
	Road safety audit	14	9	5	5
	GIS and road traffic accidents	12	8	8	8
	GIS and road safety audit	10	6	4	4
PubMed	Traffic accidents	10	7	6	6
	Road safety audit	9	7	5	5
	GIS and road traffic accident	13	11	6	5
	GIS and road safety audit	13	7	4	4
	Total	184	129	79	75

a high number of crashes, they could be disqualified from driving for further safety (Alsahfi, 2024).

Geographic Information Systems (GIS) also play a vital role in analyzing road traffic accidents as they provide spatial data to determine blackspots, trends, and causative determinants of accidents (Berhanu et al., 2023). GIS provides researchers and policymakers a method of synthesizing various datasets, including accident reports, road features, traffic density, and climate, in the form of maps, contrasting the distribution of accidents in the spatial and temporal domains. This helps transport authorities implement particular safety measures, such as improved road signs, traffic control, and infrastructure upgrades to reduce accident risks (Khallef et al., 2024).

Identification of blackspots is arguably the most significant application of GIS in traffic accident analysis (Berhanu et al., 2023). Using spatial analysis techniques like KDE, GIS can identify high-risk areas where accidents occur most frequently (Bhele et al., 2024). Such data assist the authorities in prioritizing interventions, such as increased police presence, road modifications, or traffic control. GIS also makes it easier to conduct spatiotemporal analysis, which enables researchers to determine accident trends depending on different times of day, days of the week, or seasonal changes, which is important for successful traffic management planning (Agoylo, 2024).

GIS enables analysts to conduct sophisticated statistical analyses of crashes (Munasinghe, 2023). Statistic: it is a GIS spatial statistical analysis software used in various fields of study, namely crash statistics, to identify

high-occurrence or high-risk blackspots through spatial correlation and distance that reflects local spatial dependence clusters more than global statistics can manage (Berhanu et al., 2023; Tola et al., 2021; ZahranEl-Said et al., 2019). The researcher's statistic (ZahranEl-Said et al., 2019) is used for the identification of accident blackspots, which assigns a high index value to clusters of blackspots and a low index value to clusters with lower accidents (Berhane G/Yesus, 2015; Loo, 2006; ZahranEl-Said et al., 2019). Despite this, the integration of crash rate analysis with spatial statistical analysis (Agoylo, 2024; Zakeer et al., 2025) improves the mapping of high-risk zones for road accidents. The method has been applied by a number of studies, including (Wang et al., 2021), which mapped blackspots for accidents in China.

Furthermore, GIS facilitates the analysis of accident causation by correlating accident data with environmental as well as infrastructural factors (Agoylo, 2024; Ashraf et al., 2025). For example, it can explore the relationship between accident rate and road curvature, speed limit, illumination, or pedestrian presence. Such a unified view supports data driven decision making, allowing transportation planners to design safer road networks. As GIS technology improves, real time traffic data, artificial intelligence, and machine learning models can also be used to improve accident prediction and prevention strategies, ultimately promoting road safety. Table 3, shows the synthesis matrix of GIS techniques for road traffic accident (RTA) and road safety audit (RSA).

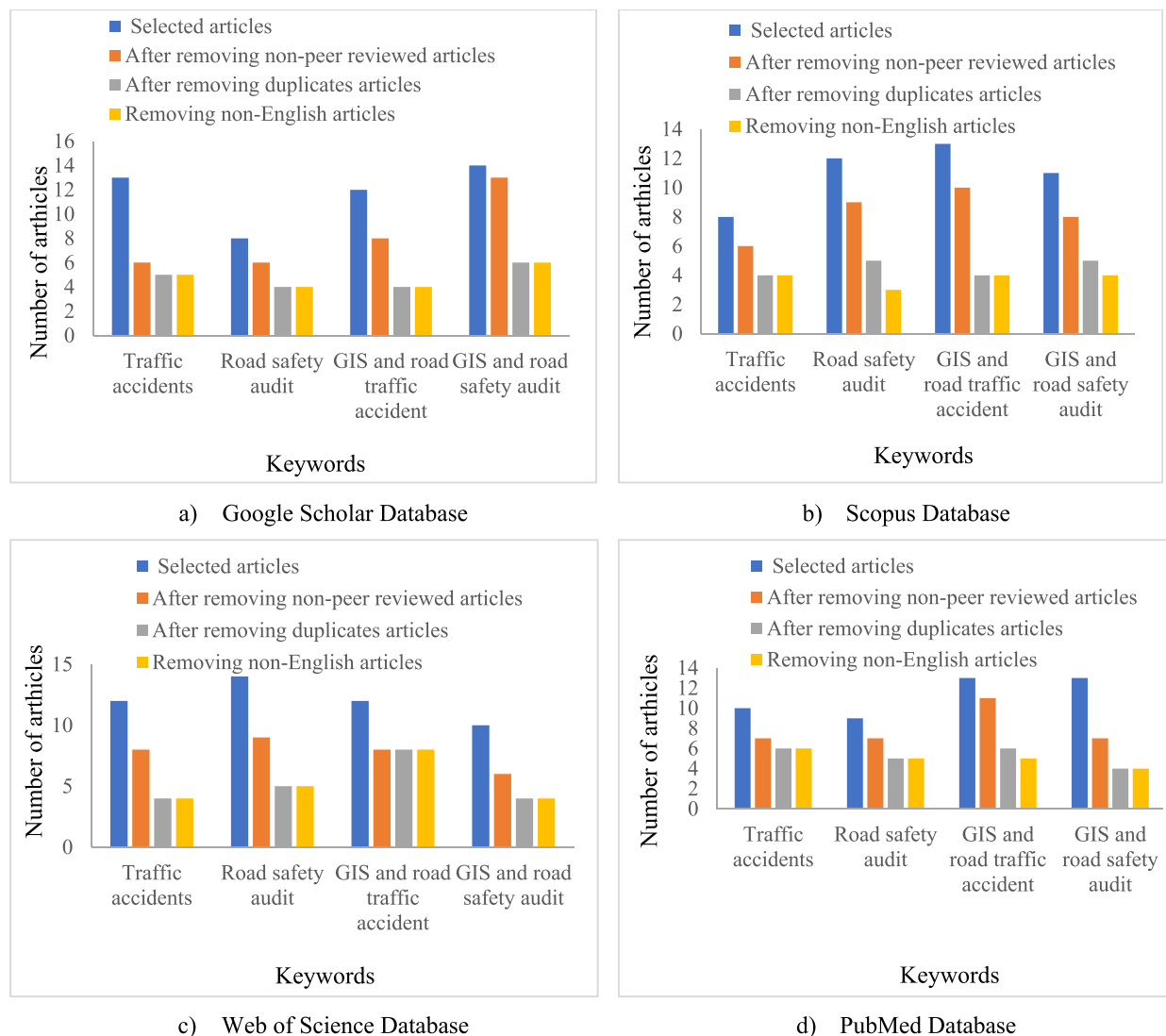


Fig. 2 Keyword-by-database retrieval volumes and screening outcomes

2.1.2 Network safety, exposure, and severity modeling

Three general methods most commonly employed to analyze accident blackspot locations are crash rate analysis, KDE, and Getis-Ord-Gi* spatial analysis. KDE is a density estimation method to graphically represent clusters, and Getis-Ord-Gi* measures the analysis of spatial autocorrelation in order for future blackspots to occur. Crash rate, however, analyzes accidents in terms of consideration of rates with respect to measures of exposure. These techniques are complementary sources of information about an accident. Spatial distribution, with its own relative strengths and weaknesses (Munasinghe, 2023; Turki et al., 2022; ZahranEl-Said et al., 2019). Researchers of the paper said that the use of Getis-Ord-Gi* spatial statistics analysis and crash rate analysis techniques allows us to identify these spatial

sets of accidents that couldn't be identified solely based on crash rates, makes accident risk more varied, provides a more complete description, enables cross-validation of results against one another, and thus enables evidence-based enhancement of road safety.

In general, the safety of road transportation networks is of utmost importance, as it is vital to every activity in human life; consistency, network operation, predictability, traffic, and road environment all affect the performance of safety of road transportation networks (Tola et al., 2021). There is curiosity among city planners and traffic engineers regarding various road network structures and what they portend for traffic safety. Utilization of GIS and GIS Road Safety Analysis Tools, such as the prediction of accident prone locations, has

Table 3 Synthesis matrix of GIS techniques for Road Traffic Accident (RTA) and Road Safety Audit (RSA)

Technique	Primary purpose	Typical spatial unit	Core inputs	Strengths	Limitations	When to prefer	Author
KDE	- Identify continuous accident density "blackspots"	- Road segments, intersections, grid cells	- Crash point locations (x,y), counts	- Simple, visual, and effective in dense urban networks	- Sensitive to bandwidth choice; no statistical test	- Initial blackspot detection in urban areas	(Bhele et al., 2024; Wang et al., 2021)
Getis-Ord Gi ^a	- Detect statistically significant hot/cold spots	- Grid cells, zones	- Crash counts; spatial weights	- Provides statistical inference; controls false positives	- Requires defined spatial weights; sensitive to MAUP ^a	- Confirming KDE blackspots with significance	(Munasinghe, 2023; Turki et al., 2022; ZahranEl-Said et al., 2019)
Crash-rate analysis	- Normalize accidents by exposure (traffic volume, length)	- Road segments	- Crash counts; traffic flow (AADT); segment length	- Easy to compute; accounts for exposure	- Regression-to-the mean; unstable in low-volume sites	- Screening roads with exposure measures available	(Mansuri et al., 2015)
Empirical Bayes (EB)	- Stabilize crash risk estimates	- Road segments, intersections	- Crash counts; traffic flow, historical averages	- Corrects regression to the mean; reliable in low-count data	- Requires reference population; complex	- Prioritizing treatment sites with low crash counts	(Erdogan et al., 2015; Tola et al., 2021; Zakeer et al., 2025)
k-means clustering	- Partition crashes into groups	- Road network segments	- Crash attributes (location, time, severity)	- Simple; fast; interpretable	- Requires predefined k; spherical clusters only	- Exploratory grouping of crash patterns	(Anderson, 2009)
Accident Severity Index (ASI)	- Prioritize sites by severity/weighting	- Road segments, intersections	- Crash severity data; expert weights	- Incorporates severity & multi-criteria	- Subjective weighting; limited comparability	- Ranking blackspots by severity	(Bilaşco & Man, 2024; Zakeer et al., 2025)

^a MAUP Modifiable Areal Unit Problem

Table 4 Summary of application of GIS in road traffic accidents analysis and road safety audit studies

Paper Title	Technique Used	Key Features	Findings	Limitations	Autor
Spatio-Temporal Assessment of Road Traffic Accidents in Mekelle City	GIS, ArcGIS 9, KDE, Statistical Analysis (SPSS)	- Uses GIS to analyze accident blackspots over four years (2008–2011) - Identifies blackspots and high-frequency accident areas - Assesses the socio-economic impacts of accidents	- 1275 accidents occurred in four years, and 34 accident blackspots were identified - Traffic accidents and casualties are increasing, requiring urgent intervention	- Missing or incomplete accident data due to police reporting limitations - Approximate accident locations are used due to a lack of GPS data	(Girmay Giday Kindaya & Hishe, 2014)
GIS-Based Traffic Accident Study in Japan	GIS, Microcomputer Accident Analysis Package, Cluster Analysis, Density Analysis	- Focus on municipal roads using hazardous spot and hazardous mesh indices - Introduces Accident Hazardous Spot and Accident Hazardous Mesh	- Hazardous meshes are concentrated in the city center - Traffic accidents on municipal roads need different measures compared to national roads	- Limited to a single city - Requires high-quality GIS data - Does not consider the behavioral aspects of drivers	(Funamoto et al., 2010)
GIS-Based Road Safety Audit	GIS-based software, Digitization, Vector Analysis	- Uses GIS to simplify road safety audits - Digitizes road features such as superelevation and signage placement - Helps in identifying accident prone locations for early intervention	- GIS-based road audits can highlight accident-prone locations early - Can improve road safety measures if integrated with GPS and Remote Sensing	- Focuses on a 3 km road segment, limiting generalizability - Does not integrate real-time traffic data	(Sayed & Mhaske, 2013)

made it possible to analyze traffic safety. Spatial statistics like Getis-Ord statistics are powerful spatial cluster algorithms for identifying accident spatial patterns and blackspots (Katanalp & Eren, 2022; Ma et al., 2021). However, the integration of crash frequency analysis and Getis-Ord spatial statistical analysis tools can identify blackspots of high risk road traffic accidents, such as spatial clusters of high risk road traffic accidents undetectable through the assistance of individual crash rates, and thus tailor interventions towards enhancing road safety (Agoylo, 2024; Tola et al., 2021). Table 4, shows the overview of the application of GIS in road traffic accident analysis and road safety audit studies.

2.2 GIS-based accident blackspot identification and clustering method

Road traffic accidents are a great threat to public safety and economic stability (Bhele et al., 2024; Bilaşco & Man, 2024; Erdogan et al., 2015; Khallef et al., 2024). Blackspots, high risk locations with large accident numbers, should be correctly identified to enable effective road safety planning (T. Ashraf et al., 2025; Science et al., 2022; Zakeer et al., 2025). GIS provides an excellent tool for the analysis of accident data, identification of blackspots, and the implementation of targeted safety measures (Ordjević, 2024). Commonly used GIS-based methods include KDE in spatial analysis (Bhele et al., 2024; Munasinghe, 2023), blackspot frequency detection (Zahran et al., 2021), and ASI in severity priority (Zakeer et al., 2025). More advanced cluster algorithms, including K-means and DBSCAN, further delineate accident hazard zone mapping from isolated cases (Anderson, 2009).

Using GIS to analyze crash data, transport planners can identify accident trends and implement evidence-based safety measures such as upgraded road infrastructure, speed management, and traffic-control programs. Integrating clustering algorithms improves the precision of blackspot detection, enabling agencies to focus on critical areas with tailored interventions (Agoylo, 2024; Turki et al., 2022). The review of existing literature demonstrates the effectiveness of the GIS approach.

Blackspots were delineated and validated using spatial analysis techniques that highlighted locations with recurrent crashes (Berhanu et al., 2023; Harirforoush & Bellalite, 2019). Clustering analysis further differentiated these sites and grouped them into classes with shared attributes such as high-speed zones or intersections offering clues to underlying causal factors (Ma et al., 2021). Practical recommendations for road safety improvement include proper signage, reduced speed limits, and enhanced road design (Djalante, 2020).

2.3 Comparative analysis of different GIS-based approaches

GIS has played a central role in identifying blackspots by enabling precise spatial analysis that supports accident reduction. A variety of GIS-based methods have been developed for this purpose, each with its own strengths and limitations. Observational comparisons indicate that spatial-autocorrelation techniques better capture the distribution of road-traffic accidents especially when accidents are unevenly distributed across the network (Berhanu et al., 2023; Bhele et al., 2024). Table 5 presents a comparative analysis of GIS-based methods for spatiotemporal analysis and blackspot identification.

Kernel density estimation (KDE) is the most commonly employed technique, producing a smooth surface of accident density across a region. It is particularly well suited to urban settings where accidents cluster spatially (Bhele et al., 2024; Munasinghe, 2023). Reviews also highlight that combining KDE with spatial-autocorrelation measures improves the precision of blackspot identification by accounting for both accident intensity and spatial arrangement (Bhele et al., 2024; Harirforoush & Bellalite, 2019; Wang et al., 2021).

The Empirical Bayes (EB) method has gained prominence because it corrects regression-to-the-mean bias by incorporating historical accident data and traffic flow (Tola et al., 2021; Zakeer et al., 2025). Detailed analyses show that EB improves the accuracy and reliability of blackspot identification relative to traditional methods (Erdogan et al., 2015; Turki et al., 2022). In summary, KDE and EB provide more robust analyses by accounting for spatial patterns and statistical bias. Selecting an appropriate methodology should reflect the characteristics of the study area and the available data to effectively identify and mitigate accident blackspots.

2.4 Effectiveness of GIS in road traffic safety analysis

Transport planning carries significant safety implications, as emphasized by a critical review of 35 major metropolitan areas (Berhanu et al., 2023). Ensuring a safe road network is imperative because it affects many dimensions of daily life. Road-traffic safety performance depends on a broad set of factors including traffic flow, network operations, predictability, consistency, and the surrounding road environment (Katanalp & Eren, 2022; Zahran et al., 2021). Greater focus on motorbike and car safety is putting road users and road safety practitioners around the globe in the spotlight, with specific regard to developing countries (Katanalp & Eren, 2022). Urban planners and transport engineers are highly sensitive to the traffic safety implications of imbalanced road network designs, with a principal platform for effective intervention

Table 5 Comparative analysis of GIS-based methods for spatial temporal analysis and road traffic accidents blackspot identification

Application of GIS	GIS-Based Blackspot Identification	Methodology	Application	Outcome	Autor
- GIS is effective in identifying accident blackspots and clustering accidents - Useful for statistical analysis & visualization of accident data	- Uses KDE and Getis-Ord Gi* for blackspot analysis	- Statistical analysis, blackspot mapping, and clustering	- Safety in dense urban areas	- Identification of high-risk areas and recommendations for traffic law enforcement	(Aati et al., 2024)
- GIS is effective in spatial analysis and identifying accident-prone areas - Useful for visualizing the relationship between population movement & accidents	- Uses standard distance analysis and density analysis to identify accident prone areas in urban areas	- Spatial and statistical analysis using GIS	- Road safety in urban areas	- Identification of high risk urban areas with corresponding traffic management strategies	(Alsaifi, 2024; Bourrelly, 2015)
- GIS is effective in identifying high risk road traffic accidents and proposing targeted interventions	- Uses KDE and spatial autocorrelation for blackspot analysis	- Spatial analysis using GIS and statistical tools	- Road safety in urban and rural areas	- Identify high risk areas & recommendations for infrastructure improvements	(Nayak & Goyal, 2024)
- GIS is effective in identifying accident blackspots and temporal patterns	- Uses KDE for blackspot analysis	- Spatio temporal analysis using GIS and statistical tools	- Road safety in urban and rural areas	- Identification of high risk areas and recommendations for road safety interventions	(Bhele et al., 2024; Turki et al., 2022)

(Berhanu et al., 2023). Urbanity and the nature of spatial development greatly shape traffic safety policy, striving for sustainable equilibrium in terms of requirements, opportunities, and road quality (Djalante, 2020).

A basic indicator of quantifying road safety is introduced through the analysis of costs to society caused by accidents and injuries (Sayed & Mhaske, 2013). The severity of the crashes depends primarily on the highest severity of the injured participant, provoking a systematic method for authorities to focus on road safety improvement for specific road sections (Funamoto et al., 2010; Satria, 2020). Identification of traffic accident risk cluster areas stands as a crucial starting point toward fostering traffic safety (Mansuri et al., 2015). By such a paradigm, blackspot strategies subsequently become important as instrumental devices that facilitate the transposition of low-cost road safety interventions to counter traffic crashes (Tola et al., 2021).

3 Research gaps and future directions

Despite the growing body of work on GIS-based road traffic accident analysis and road safety audits, several important gaps remain. First, there is a clear regional disparity in the literature, with most studies concentrated in high-income urban areas, while low-income regions and secondary road networks remain largely underrepresented. This limits the global applicability of current findings and underscores the need for open crash registries and standardized geo-coding practices to ensure broader coverage. Second, the integration of diverse data sources remains limited. Very few studies utilize real-time data streams such as loop detectors, connected vehicle basic safety messages, or crowdsourced traffic reports from platforms like Waze and Twitter. Moreover, behavioral and exposure-related variables such as driver speed choice, mobile phone distraction, and pedestrian or cyclist volumes are rarely included, restricting the ability to model human factors in crash risk. Third, methodological gaps persist, particularly the lack of validation of blackspot stability over time and the limited number of studies that systematically compare different GIS-based techniques against the same ground-truth dataset. Addressing these shortcomings requires novel approaches: the incorporation of Internet of Things (IoT) sensor data into GIS platforms, the application of deep learning methods such as graph neural networks for spatiotemporal crash forecasting, and the use of crowdsourced or citizen-reported hazard data with robust outlier detection mechanisms. Finally, reproducibility must be prioritized through the open sharing of codebooks, analytical workflows, and geospatial datasets, enabling consistent benchmarking and accelerating advances in GIS-based road safety research.

4 Conclusion

This systematic review demonstrates that road safety remains a pressing global issue, as road traffic accidents continue to claim lives and disrupt communities worldwide. GIS has proven to be a valuable tool for analyzing accident data, enabling blackspot detection, spatiotemporal trend analysis, and integration of multiple data layers to inform targeted interventions. Techniques such as KDE, Getis–Ord G_i^* , crash rate analysis, and Empirical Bayes have been widely applied, offering both visual and statistically robust insights into accident patterns.

Beyond methodological contributions, GIS applications have strengthened evidence-based policymaking by assisting transport authorities, engineers, and planners in prioritizing high-risk locations and designing safer transport systems. However, challenges persist, including uneven data coverage, methodological inconsistency, and limited use of behavioral, exposure, and real-time datasets.

Looking forward, future research should focus on:

- Integration of IoT and real-time traffic data to enhance predictive modeling and proactive safety management.
- Application of deep learning and advanced AI models to capture complex spatiotemporal crash dynamics.
- Incorporation of citizen-reported and crowdsourced data to complement traditional crash databases and improve situational awareness.
- Standardize GIS methods and share data openly to ensure reproducibility and enable cross-regional comparisons.

By addressing these gaps, GIS can evolve from a descriptive analytical tool into a comprehensive decision-support platform that empowers policymakers and practitioners to implement timely, data-driven interventions. Ultimately, this will not only improve road safety outcomes but also contribute to broader public health, sustainable mobility, and equitable transport systems.

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Authors' contributions

Destaw Kifle Endashaw is a PhD student in the Department of Civil & Environmental Engineering, King Fahd University of Petroleum and Minerals (KFUPM). He contributed to the study's conceptualization, writing, and revision, ensuring clarity and quality in response to reviewer comments. Kidanemariam Alula Habtegiorgis is an MSc student in the Department of Civil & Environmental Engineering, KFUPM, and he contributed by critically proofreading and editing the manuscript in response to reviewer comments, ensuring its clarity and quality. Baqer Mohammad Al-Ramadan is a Contributing Professor in the Department of Architecture & City Design, KFUPM. He served as the supervisor

for the research and provided editorial support. Hassan Musaed Al-Ahmadi is a Professor in the Department of Civil & Environmental Engineering, College of Design and Built Environment, KFUPM. He supervised the research process, contributed to the conceptualization, provided editorial support, critically reviewed the manuscript, and refined its content. Basha Fayissa Deressa is a PhD student in the Department of Civil & Environmental Engineering, KFUPM, and he contributed by editing the document. This paper, titled "A Systematic Review on GIS-Based Road Traffic Accident Analysis and Road Safety Audit," supports advancements in transportation infrastructure and road safety by providing a comprehensive overview of how Geographic Information Systems (GIS) are used to analyze traffic accidents and improve safety audits.

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Data availability

Not applicable for this section.

Declarations

Competing interests

The author declares no conflict of interest.

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